MSc

Course title: Bayes Methods
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1. Let $A = [0, 1]^2$ be the unit square region of $\Re^2$. For $i = 1, ..., m$, let $y_i = (y_{i,1}, y_{i,2})$ be points in A and let $y = \{y_1, ..., y_m\}$ be a set of m points. The number and locations of the points are distributed according to a Strauss point process with parameters $\lambda > 0, 0 \leq \gamma \leq 1$ and $R \geq 0$. In particular, let $c(y; R)$ be the function

$$c(y; R) = \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} \mathbb{I}_{|y_i - y_j| < R}$$

counting the number of R-close pairs of points in y. Let Ω_m be the set of all distinguishable sets of m points $y = \{y_1, ..., y_m\}$ in $y \subset A$ and let $\Omega = \cup_{m=0}^{\infty} \Omega_m$. The density of $y|\lambda, \gamma, R$ is

$$p(y|\lambda, \gamma, R) = \xi(\lambda, \gamma, R)\gamma^{c(y,R)}\lambda^m,$$

where $\xi(\lambda, \gamma, R)$ is a normalising constant ensuring the density is normalised to one over Ω.

(a) [8 marks] Suppose y is observed and we wish to estimate λ, γ, R.

(i) Write down the posterior distribution $\pi(\lambda, \gamma, R|y)$ in as much detail as you can, but without specifying priors for λ, γ, R.

Solution: (Bookwork - from a practical)

$$\pi(\lambda, \gamma, R|y) \propto \xi(\lambda, \gamma, R)\gamma^{c(y,R)}\lambda^m \pi(\lambda, \gamma, R).$$

(ii) Explain why $\pi(\lambda, \gamma, R|y)$ is doubly intractable.

Solution: (Simple application similar to PS/BW) Posterior $p(\theta|y)$ is doubly intractable if we cannot evaluate $p(y|\theta)\pi(\theta)$. This typically occurs because the likelihood cannot be evaluated as a function of θ. For the parameters of a Strauss process, the posterior $\pi(\lambda, \gamma, R|y)$ is an intractable function of λ, γ, R. To see this, observe that we need to be able to evaluate $\xi(\lambda, \gamma, R)$ in order to evaluate $\pi(\lambda, \gamma, R|y)$ even up to an overall constant. However

$$\xi(\lambda, \gamma, R) = \sum_{n=0}^{\infty} \int_{\Omega_m} \gamma^{c(z,R)}\lambda^n d^{2n}z.$$

This integral is intractable so we cannot evaluate $\pi(\lambda, \gamma, R|y)$ up to a constant.

(iii) Suppose an algorithm simulating $x \sim p(x|\lambda, \gamma, R)$ is available. Give an ABC algorithm approximately targeting $\pi(\lambda, \gamma, R|y)$ (state what summary statistics and distance measure you are using).

Solution: (Bookwork - from a practical) Summary statistics $S(x) = (c(x; R), m(x))$, the number of R-close pairs of points and the number of points, are a natural starting point. These are clearly sufficient for γ and λ if R is known. We would do better to seek a third statistic, but this is acceptable (since the prior also informs the parameters). A straightforward choice of distance would be $d(S(x), S(y))^2 = (S(x) - S(y))^T(S(x) - S(y))$ though we might do better to scale $c(x; R)$ and $m(x)$ so that they have the same scale of variation. Let a scale $\rho > 0$ be given. Our ABC algorithm returning one sample is

1) Simulate $\lambda, \gamma, R \sim \pi(\lambda, \gamma, R)$ and $x \sim p(x|\lambda, \gamma, R)$.

2) If $d(S(x), S(y)) < \rho$ return λ, γ, R as an approximate posterior sample, and otherwise goto step 1.

We should experiment with different values of ρ, reducing ρ till, for example, the results for ρ and $\rho/2$ are similar.

(b) [12 marks] Suppose now λ, γ and R are known, but the point pattern y is in fact a thinned realisation of a Strauss process. The true underlying set of points $z = \{z_1, ..., z_n\}, z \in \Omega$, $n \geq m$ has a Strauss distribution in $A = [0, 1]^2$,

$$p(z|\lambda, \gamma, R) \propto \gamma^{c(z,R)} \lambda^n.$$

For $i = 1, ..., n$, point z_i is observed independently, so $z_i \in y$, with probability q, and otherwise z_i is missed. In that case $z_i \notin y$. Here q, the probability a point gets observed, is unknown with prior $\pi(q)$ independent of z. We wish to estimate a credible interval for q.

(i) Let z be a point pattern of n points satisfying $y \subseteq z$. Calculate $\pi(z, q|y)$

Solution: (new - similar to practical) We have $p(y|z, q) = q^m(1 - q)^{n-m}$ and $\pi(z, q|y) \propto p(y|z, q)\pi(z, q)$ with $\pi(z, q) = p(z|\lambda, \gamma, R)\pi(q)$ so

$$\pi(z, q|y) \propto q^m(1 - q)^{n-m}\gamma^{c(z,R)}\lambda^n\pi(q).$$

(ii) Give a reversible jump MCMC algorithm targeting $\pi(z, q|y)$. Your algorithm should contain updates for adding and deleting points $(x_1, x_2) \in z \setminus y$, and an update for q. Give the acceptance probabilities as explicitly as you can.

Solution: (new - similar to practical) Suppose $X_t = (z, q)$ with $z = (y_1, ..., y_m, z_{m+1}, ..., z_n)$ and $n \geq 0$. The updates generating candidate states are as follows.

1. To add a point throw a point x UAR in A. Let $z' = z \cup x$ and $n' = n + 1$. The number of R-close pairs goes up to $c(z'; R)$ and the number of points goes up to n'.
2. To delete a point, choose a point $z_i \in z \setminus y$ at random from $i = m+1, ..., n$ (and if $z \setminus y$ is empty then do nothing, ie set $X_{t+1} = X_t$). Let $z' = z \setminus z_i$ and $n' = n - 1$. The number of R-close pairs goes down to $c(z')$ and the number of points goes down to n'.
3. To update q sample $q' \sim U(0, 1)$.

Choose add, delete and update-q moves with probability 1/3, ie set $p_{n,n+1} = p_{n,n-1} = p_{n,n} = 1/3$ below. The acceptance probability to add a point is

$$\begin{aligned} \alpha(z', q|z, q) &= \min\left\{1, \frac{\pi(z', q|y)}{\pi(z, q|y)} \frac{p_{n,n'}Q(z|z')}{p_{n',n}Q(z'|z)}\right\} \\ &= \min\left\{1, \frac{q^m(1 - q)^{n+1-m}\gamma^{c(z',R)}\lambda^{n+1}\pi(q)}{q^m(1 - q)^{n-m}\gamma^{c(z,R)}\lambda^n\pi(q)} \frac{1/(n - m + 1)}{1}\right\}. \end{aligned}$$

since $Q(z'|z) = Q(x) = U(A)$ is the density of the new point which is uniform on A, the unit square, and $Q(z'|z) = 1/(n-m+1)$ since there are $n-m+1$ points to choose from when we make the reversing deletion. Cancelling,

$$\alpha(z', q|z, q) = \min\left\{1,\ \frac{(1-q)\lambda\gamma^{c(z';R)-c(z;R)}}{n-m+1}\right\}$$

The acceptance probability for deleting a point is

$$\begin{aligned}\alpha(z', q|z, q) &= \min\left\{1,\ \frac{\pi(z', q|y)}{\pi(z, q|y)}\frac{p_{n,n'}Q(z|z')}{p_{n',n}Q(z'|z)}\right\}\\ &= \min\left\{1,\ \frac{q^m(1-q)^{n-1-m}\gamma^{c(z',R)}\lambda^{n-1}\pi(q)}{q^m(1-q)^{n-m}\gamma^{c(z,R)}\lambda^n\pi(q)}\frac{1}{1/(n-m)}\right\}.\end{aligned}$$

For this deletion move $Q(z'|z) = 1/(n-m)$ in the expression above since the deleted point is chosen UAR from the points in z which are not in y. Cancelling,

$$\alpha(z', q|z, q) = \min\left\{1,\ \frac{(n-m)\gamma^{c(z';R)-c(z;R)}}{(1-q)\lambda}\right\}$$

The acceptance probability for q' (where $n' = n$ as z doesnt change) is

$$\begin{aligned}\alpha(z, q'|z, q) &= \min\left\{1,\ \frac{\pi(z, q'|y)}{\pi(z, q|y)}\frac{p_{n,n}Q(q|q')}{p_{n,n}Q(q'|q)}\right\}\\ &= \min\left\{1,\ \frac{q'^m(1-q')^{n-m}\gamma^{c(z,R)}\lambda^n\pi(q')}{q^m(1-q)^{n-m}\gamma^{c(z,R)}\lambda^n\pi(q)}\right\}.\\ &= \min\left\{1,\ \frac{q'^m(1-q')^{n-m}\pi(q')}{q^m(1-q)^{n-m}\pi(q)}\right\}.\end{aligned}$$

(iii) Explain how to compute a $100(1-\alpha)\%$ equal tailed credible interval for q from MCMC output $q^{(t)}, t = 1, ..., T$. What is the relationship to model averaging here?

Solution: (new) To get an equal tailed interval sort the $q^{(t)}$ values to get $q_{(t)}, t = 1, ..., T$ say. Then if $t_1 = \lfloor \alpha T \rfloor$ and $t_2 = \lceil (1-\alpha)T \rceil$ then $[q_{(t_1)}, q_{(t_2)}]$ is an approximate $\%100(1-\alpha)$ credible interval for $q|y$. This is a credible interval for

$$\pi(q|y) = \sum_{n=m}^{\infty} \int_{\Omega_{n-m}} \pi(z, q|y) dz_{m+1}...dz_n.$$

This distribution is averaged over models (indexed by n) and parameter values within models (the unknown $z_{m+1}...z_n$).

2. (a) [5 marks] A random distribution $G \sim DP(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H on Ω.

(i) Give a definition of a Dirichlet Process.

> **Solution:** (BW) $G \sim DP(\alpha, H)$ iff $\forall$ partitions $A_1, A_2, ..., A_r$ of Ω
>
> $$G(A_1), ..., G(A_r) \sim \mathsf{Dirichlet}(\alpha H(A_1), ..., \alpha H(A_r))$$

(ii) Calculate the expectation and variance of $G(A), A \subseteq \Omega$ in terms of α and $H(A)$, and briefly interpret the role α and H play in determining the distribution of G.

Hint: If $X \sim$*Beta*(a, b), *then* $E(X) = \dfrac{a}{a+b}$, *and* $\mathrm{var}(X) = \dfrac{ab}{(a+b)^2(a+b+1)}$.

> **Solution:** (BW) First, $G(A), G(A^c) \sim \mathrm{Beta}(a, b)$ with $a = \alpha H(A)$ and $b = \alpha H(A^c)$ by the agglomerative property of Dirichlet distributions. We have
>
> $$E(G(A)) = \frac{a}{a+b} = \frac{\alpha H(A)}{\alpha H(A) + \alpha H(A^c)} = H(A)$$
>
> since $H(A) + H(A^c) = 1$. The variance is
>
> $$\begin{aligned} \mathrm{var}(G(A)) &= \frac{ab}{(a+b)^2(a+b+1)} \\ &= \frac{\alpha^2 H(A)H(A^c)}{(\alpha H(A) + \alpha H(A^c))^2(\alpha H(A) + \alpha H(A^c) + 1)} \\ &= \frac{H(A)(1 - H(A))}{\alpha + 1} \end{aligned}$$
>
> Here H is the "centre" of the distribution of G in the sense that it is the mean about which G varies, and α controls the variance of G around H. When α is large, G is similar in distribution to H.

(b) [9 marks] Consider regression with a Dirichlet-Normal mixture for the error distribution. Let $G \sim DP(\alpha, H)$ with $H(d\theta_i) = N(d\theta_i; 0, \sigma_\theta^2)$. Here $\theta_i \in \Re$ so $\Omega = \Re$. For $i = 1, ..., n$ let x_i, y_i be a scalar covariate and response and β a scalar effect. The regression is

$$y_i = \theta_i + x_i\beta + \epsilon_i, \qquad \text{with } \epsilon_i \sim N(0, \sigma_\epsilon^2) \text{ and } \theta_i \sim G$$

independently for each $i = 1, ..., n$. Let $\theta = (\theta_1, ..., \theta_n)$ and $\sigma = (\sigma_\epsilon, \sigma_\theta)$.

(i) Let $\theta^* = (\theta_1^*, ..., \theta_K^*)$ list the distinct values in θ. Let $S = (S_1, ..., S_K)$ be the partition of $\{1, 2, ..., n\}$ satisfying $i \in S_k$ if $\theta_i = \theta_k^*$ and let $n_k = \mathrm{card}(S_k)$. Let $P(S)$ be the probability θ has partition S.
Assume a prior $\pi(\beta, \sigma)$ is available. Write down the posterior for β, σ, θ^* and S in as much detail as you can (α is known and fixed).

Hint: You may use without proof that $P(S) = \dfrac{\alpha^K \prod_{k=1}^K (n_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)}$.

Solution: (variation on lecture and problem sheet example)

$$\pi(\beta,\sigma,\theta^*,S|y) \propto \pi(\beta,\sigma)P(S)\pi(\theta^*|S,\sigma_\theta)L(\theta^*,\beta,\sigma_\epsilon;y)$$

Here $\pi(\beta,\sigma)$ is given, $P(S)$ is given in the hint,

$$\pi(\theta^*|S,\sigma_\theta) = \prod_{k=1}^{K}\prod_{i\in S_k} N(\theta^*_k;0,\sigma^2_\theta),$$

and

$$L(\theta^*,\beta,\sigma_\epsilon;y) = \prod_{k=1}^{K}\prod_{i\in S_k} N(y_i;\theta^*_k+x_i\beta,\sigma^2_\epsilon).$$

(ii) Let K^{-i}, $S^{-i}=(S^{-i}_1,...,S^{-i}_{K^{-i}})$ and θ^{*-i} correspond to K, $S=(S_1,...,S_K)$ and θ^* after i is removed from its partition set (but note that y_i is retained). Calculate $\Pr(i\in S_k|S^{-i},\beta,\sigma,\theta^{*-i},y)$, the conditional distribution for the partition for the i'th response, given the other parameters and the partition for the rest of the data and briefly indicate why this distribution is useful as part of an MCMC algorithm targeting $\pi(\beta,\sigma,\theta^*,S|y)$.

Solution: (variation on PS problem/lecture) Let $|n^{-i}_k| = \text{card}(S^{-i}_k)$. The full posterior is

$$\pi(\beta,\sigma,\theta^*,S|y) \propto \pi(\beta,\sigma)\prod_{k=1}\alpha(n_k-1)!N(\theta^*_k;0,\sigma^2_\theta)\prod_{i\in S_k} N(y_i;\theta^*_k+x_i\beta,\sigma^2_\epsilon).$$

If k is in $1,...,K^{-i}$, so i is going into an existing cluster, then

$$\pi(\beta,\sigma,\theta^{*-i},S^{-i}|y) \propto \pi(\beta,\sigma)\prod_{k=1}^{K^{-i}}\alpha(n^{-i}_k-1)!N({\theta^*}^{-i}_k;0,\sigma^2_\theta)\prod_{j\in S_k,j\neq i} N(y_j;{\theta^*}^{-i}_k+x_j\beta,\sigma^2_\epsilon).$$

If $k=K^{-i}+1$, so k is going into a new cluster, then

$$\pi(\beta,\sigma,\theta^{*-i},S^{-i}|y) \propto \pi(\beta,\sigma)\prod_{k=1}^{K^{-i}+1}\alpha(n^{-i}_k-1)!N({\theta^*}^{-i}_k;0,\sigma^2_\theta)\prod_{j\in S_k,j\neq i} N(y_j;{\theta^*}^{-i}_k+x_j\beta,\sigma^2_\epsilon).$$

The conditional we want is

$$\Pr(i\in S_k|S^{-i},\beta,\sigma,\theta^{*-i},y) = \frac{\pi(\beta,\sigma,\theta^*,S|y)}{\pi(\beta,\sigma,\theta^{*-i},S^{-i}|y)},$$

so looking at the different terms we see

$$\Pr(i\in S_k|S^{-i},\theta^{*-i},y) = \begin{cases} |n^{-i}_k|N(y_i;{\theta^*}^{-i}_k+x_i\beta,\sigma^2_\epsilon) & \text{if } k\in\{1,2,...,K^{-i}\} \\ \alpha N(y_i;{\theta^*}_{K^{-i}+1}+x_i\beta,\sigma^2_\epsilon) & \text{if } k=K^{-i}+1. \end{cases}$$

This is the distribution we need to implement a Gibbs sampler for the partition, S. It would be accompanied by MH-MCMC updates for the other parameters.

(c) [6 marks]

(i) Calculate a Laplace approximation to $\log(p(y|S))$, where $p(y|S)$ is the marginal likelihood for the data y given a partition S.
Hint: consider $z = (\beta, \sigma, \theta^*)$ *in the Laplace approximation,*

$$\int e^{-nh(z)} dz = e^{nh(z)} (2\pi/n)^{p/2} |\Sigma|^{1/2} + O(n^{-3/2})$$

where $h(z)$ *is a given smooth function of* $z = (z_1, ..., z_p)$, $\Sigma^{-1} = \left. \frac{\partial^2 h}{\partial z \partial z^T} \right|_{z=\widehat{z}}$ *is the Hessian of* $h(z)$ *at* $\widehat{z}$ *with determinant* $|\Sigma|^{-1}$, *and* $\widehat{z}$ *maximises* $h(z)$.

Solution: (new) First,

$$p(y|S) = \int \pi(\beta, \sigma) \pi(\theta^* | S, \sigma_\theta) L(\theta^*, \beta, \sigma_\epsilon; y) d\beta d^2\sigma d^K \theta^*.$$

Let $\widehat{\beta}, \widehat{\sigma}, \widehat{\theta}^*$ be the mode of

$$\pi(\beta, \sigma, \theta^* | y, S) \propto \pi(\beta, \sigma) \pi(\theta^* | S, \sigma_\theta) L(\theta^*, \beta, \sigma_\epsilon; y).$$

Let $\overline{\ell}(\theta^*, \beta, \sigma_\epsilon; y) = n^{-1} \sum_k \sum_{i \in S_k} \log(N(y_i; {\theta^*}_k + x_i \beta, \sigma_\epsilon^2))$ and

$$h(\beta, \sigma, \theta^*) = \frac{1}{n} \log(\pi(\beta, \sigma)) + \frac{1}{n} \log(\pi(\theta^* | S, \sigma_\theta)) + \overline{\ell}(\theta^*, \beta, \sigma_\epsilon; y).$$

Then

$$p(y|S) = \int e^{-nh(\beta, \sigma, \theta^*)} d\beta d^2\sigma d^K \theta^*.$$

Applying the Laplace approximation,

$$\begin{aligned} \log(p(y|S)) \quad &\simeq \quad n\overline{\ell}(\widehat{\theta}^*, \widehat{\beta}, \widehat{\sigma}_\epsilon; y) + \log(\pi(\widehat{\beta}, \widehat{\sigma})) + \log(\pi(\widehat{\theta}^* | S)) \\ &\quad + \frac{p}{2} \log(2\pi/n) - \frac{1}{2} \log(|\Sigma|). \end{aligned}$$

The question does not ask for the BIC etc, so we simply retain all these terms.

(ii) In a problem with n small it is feasible to estimate $p(y|S)$ for every possible partition S using a Laplace approximation. A researcher interested in estimating the partition does this and reports the partition with the largest estimated value of $p(y|S)$ as a best estimate for the unknown true partition. What criticisms do you have of this procedure?

Solution: It is equivalent to taking the mode of the posterior $\pi(S|y) \propto P'(S)p(y|S)$ as the estimate with $P'(S)$ the flat prior over partitions. If this prior is not a reasonable summary of existing knowledge of S then the prior will bias the estimates in a way that is not justified by any available information. For example, there is only one partition with just one group, but $2^{n-1} - 1$ partitions with two groups, so the prior probability for two groups is far higher than that for one, with no elicited justification. Secondly, the posterior probability for any single partition is typically small. Reporting a single partition seems rather pointless: it is the most probable, but still very unlikely. Thirdly, it is surely desirable to quantify the uncertainty, giving a credible set for the partition would likely be more useful. The first or last two points OK.